Unit-6: Numerical Methods

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Finite Difference

Forward Differences: The first forward difference is defined by $\Delta f(x) = f(x+h) - f(x)$

Where Δ is (Delta) called as the forward difference operator.

$\Delta^2 f(x) = \Delta[\Delta f(x)] = \Delta[f(x+h) - f(x)] = \Delta f(x+h) - \Delta f(x)$

$\Delta^2 f(x) = f(x+2h) - f(x+h) - f(x+h) + f(x) = f(x+2h) - 2f(x+h) + f(x)$

Backward Differences: The first forward difference is defined by $\nabla f(x) = f(x) - f(x-h)$

Where ∇ is (Del) called as the backward difference operator.

$\nabla^2 f(x) = \nabla[\nabla f(x)] = \nabla[f(x) - f(x-h)] = \nabla f(x) - \nabla f(x-h)$

$\nabla^2 f(x) = f(x) - f(x-h) - f(x-h) + f(x-2h) = f(x) - 2f(x-h) + f(x-2h)$

The Central Differences: The first order central difference is defined by,

$\delta f(x) = f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right)$ Where δ is called as the central difference operator.

The Shift Operator E: The shift operator is defined by,

$E^n f(x) = f(x+nh)$ $\quad$ $E^{-n} f(x) = f(x-nh)$

$E^2 f(x) = f(x+2h)$ $\quad$ $E^{-1} f(x) = f(x-h)$

$Ef(x) = f(x+h)$

The Average Operator μ: The average operator is defined by,

$$\mu f(x) = \frac{1}{2}\left[f\left(x + \frac{h}{2}\right) + f\left(x - \frac{h}{2}\right)\right]$$

Symbolic Relations

1. $E = 1 + \Delta$
2. $E\nabla = \nabla E = \Delta$
3. $(1+\Delta)(1-\nabla) = 1$
4. $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$
5. $\delta = \frac{1}{2}\left(E^{\frac{1}{2}} + E^{-\frac{1}{2}}\right)$
6. $\mu = \left[1 + \frac{\delta^2}{4}\right]^{\frac{1}{2}}$
7. $\Delta\nabla = \nabla\Delta = \delta^2$

Newton-Greogory Formula for Forward Interpolation

Consider the set of $(n+1)$ equidistant values of the function $y = f(x)$ with $y_0 = f(x_0)$,

$$y_1 = f(x_1), y_2 = f(x_2), \ldots y_n = f(x_n) \text{ and } u = \frac{x - x_0}{h}$$

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!}\Delta^4 y_0 + \cdots$$

***Note**: This formula is useful for interpolation near the **beginning** of a set of tabular values.*

Newton-Greogory Formula for Backward Interpolation

Consider the set of $(n+1)$ equidistant values of the function $y = f(x)$ with $y_0 = f(x_0)$,

$y_1 = f(x_1), y_2 = f(x_2), \dots y_n = f(x_n)$ *and* $u = \dfrac{x - x_n}{h}$

$$y = y_n + u\nabla y_n + \frac{u(u+1)}{2!}\nabla^2 y_n + \frac{u(u+1)(u+2)}{3!}\nabla^3 y_n + \frac{u(u+1)(u+2)(u+3)}{4!}\nabla^4 y_n + \cdots$$

***Note**: This formula is useful for interpolation near the **end** of a set of tabular values.*

Lagrange's Interpolation Formula

Let $y = f(x)$ be continuous and differentiable $(n+1)$ times in the interval (a, b). Given $(n+1)$ paired values are $[(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots (x_n, y_n)]$ Where $y_0 = f(x_0), y_1 = f(x_1), y_2 = f(x_2), \dots y_n = f(x_n)$. Then the Lagrange's polynomial is given by,

$y = f(x) = L_0 y_0 + L_1 y_1 + L_2 y_2 + \cdots + L_n y_n$

Where: $L_0 = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)}$, $L_1 = \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)}$, $L_2 = \frac{(x-x_0)(x-x_1)\dots(x-x_n)}{(x_2-x_0)(x_2-x_1)\dots(x_2-x_n)}$, …

- For pair of two variables: $(x_0, y_0), (x_1, y_1)$

  $y = f(x) = \frac{(x-x_1)}{(x_0-x_1)} y_0 + \frac{(x-x_0)}{(x_1-x_0)} y_1.$

- For pair of three variables: $(x_0, y_0), (x_1, y_1), (x_2, y_2)$

  $y = f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} y_2.$

Examples: Finite Difference

1. From the following table find (a) y when $x = 7$ (b) y when $x = 19$. [Ans: 5.17, 219.2]

x	8	10	12	14	16	18
$y = f(x)$	10	19	32.5	54	89.5	15.4

2. Find value of y for x = 0.5 for the following table of x, y values using Newton's forward difference formula.

x	0	1	2	3	4
y	1	5	25	100	250

3. From the following data, find the cubic polynomial and hence find $f(4)$. [Ans=41]

x	0	1	2	3
$f(x)$	1	2	1	10

4. From the tabulated values of x and y, prepare forward difference table. Find polynomial passing through the points and find its slope (derivative) at $x = 1.5$. [Ans=17.75]

x	0	2	4	6	8
y	5	29	125	341	725

5. The distance travelled by a point P in x-y plane in a mechanism is as shown in the table below. Estimate the distance travelled, velocity of point P when x = 4.5, by using Newton's backward difference formula. [Ans: 153.125]

x	1	2	3	4	5
y	14	30	62	116	198

6. Find value of y for $x = 0.5$ for the following set of values of x and y using Newton's forward difference formula. [Ans: 4.17]

x	0	1	2	3	4
y	1	5	25	100	250

Examples: Lagrange's Interpolation Formula

1. By using Lagrange's interpolation formula, Evaluate $f(5)$ given that $f(1) = 2, f(2) = 4, f(3) = 8, (4) = 16, f(7) = 128$. [Ans: 33.13]
2. Find Lagrange's polynomial passing through set of points $(0,2), (2,-2), (3,-1)$. Use it to find the $\frac{dy}{dx}$ at $x = 2$. [Ans: 0]
3. Given $(1.0)^3 = 1.000, (1.2)^3 = 1.728, (1.3)^3 = 2.197$ *and* $(1.5)^3 = 3.375$. By using Lagrange's interpolation formula, evaluate $(1.7)^3$ =? [Ans: 4.913]
4. Using Lagrange's interpolation formula to evaluate y when x=3 for the following set of values. [Ans: 9]

x	1	2	4
y	1	4	16

5. Using Lagrange's interpolating polynomial passing through set of points. Use it to find y when $x = 4$. [Ans: 6]

x	1	2	5
y	1	4	25

6. Using Lagrange's interpolating polynomial passing through set of points. Use it to find y when $x = 1.5$. [Ans: 1.75]

x	0	1	2
y	4	3	6

7. Find the form of the function y for the following data. Hence find y at $x = 4$. [Ans: 78]

x	0	1	2	5
y	2	3	12	147

Numerical Integration

Let $\int_a^b y\,dx$ where $y = f(x)$ be given for certain equidistant values of arguments say $x_0, x_0 + h$, $x_0 + 2h$, ... Let the range $b - a$ be divided into n equal parts each of width h $\therefore$ h $= \frac{b-a}{n}$.

Let $x_0 = a, x_1 = x_0 + h, x_2 = x_0 + 2h, \ldots x_n = b = x_0 + nh$

Trapezoidal Rule

$$\int_a^b y\,dx = \int_{x_0}^{x_0+nh} y\,dx = h\left[\frac{1}{2}(y_0 + y_n) + (y_1 + y_2 + \cdots + y_{n-1})\right]$$

$$\int_a^b y\,dx = h[(Mean\ of\ first\ and\ last\ ordinate) + (sum\ of\ all\ remaining\ ordinates)]$$

Simpson's $\frac{1}{3}^{rd}$ Rule

$$\int_a^b y\,dx = \int_{x_0}^{x_0+nh} y\,dx = \frac{h}{3}[(y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1}) + 2(y_2 + y_4 + \cdots + y_{n-2})]$$

$$\int_a^b y\,dx = \frac{h}{3}[(sum\ of\ first\ and\ last\ ordinate) + 4(sum\ of\ odd\ ordinates) + 2(sum\ of\ even\ ordinates)]$$

Simpson's $\frac{3}{8}^{th}$ Rule

$$\int_a^b y\,dx = \int_{x_0}^{x_0+nh} y\,dx = \frac{3h}{8}[(y_0 + y_n) + 2(y_3 + y_6 + \cdots + y_{n-3}) + 3(y_1 + y_2 + y_4 \ldots)]$$

$$\int_a^b y\,dx = \frac{3h}{8}[(sum\ of\ first\ and\ last\ ordinate) + 2(sum\ of\ multiple\ of\ 3\ ordinates) + 3(sum\ of\ remaining\ ordinates)]$$

Examples: Trapezoidal Rule, Simpson's $\frac{1}{3}^{rd}$ Rule, Simpson's $\frac{3}{8}^{th}$ Rule

1. Use Trapezoidal rule to numerically evaluate $\int_0^1 xe^{x^2}\,dx$ by taking $h = 0.1$. [Ans=0.86]
2. Find the area bounded by the curve $f(x)$ and the x $axis$ and $x = 7.47$ to $x = 7.52$ from the following data, (by Trapezoidal rule). [Ans=0.996]

x	7.47	7.48	7.49	7.50	7.51	7.52
$f(x)$	1.92	1.95	1.98	2.01	2.03	2.06

3. Evaluate $\int_0^3 \frac{dx}{1+x}$ with 7 ordinates by using Simpson's $\frac{3}{8}^{th}$ rule and hence calculate log 2. [Ans=1.388, log 2=0.6944.]
4. Evaluate $\int_0^{\frac{\pi}{2}} \frac{sinx}{x}dx$ by Simpson's $\frac{1}{3}^{rd}$ by dividing the interval into 4 parts. [Ans=1.37]
5. Evaluate $\int_1^2 \frac{1}{x}dx$ by Simpson's $\frac{1}{3}^{rd}$ with 10 intervals. [Ans=0.693067]
6. Using Simpson's $\left(\frac{3}{8}\right)^{th}$ rule, evaluate $\int_0^{\pi} \frac{\sin^2\theta}{5+4cos\theta}d\theta$ taking $h = \frac{\pi}{6}$. [Ans: 0.4021]
7. Using Simpson's $\left(\frac{1}{3}\right)^{rd}$ rule, evaluate $\int_1^2 \frac{1}{x^2}dx$ by dividing the integral into equally spaced intervals of width $h = 0.25$. [Ans: 0.5]
8. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ taking $h = \frac{1}{6}$ by using Simpson's $\frac{3}{8}^{th}$ rule. Compare the result with actual value. [Ans=0.785395, Actual value=0.7853981]

Runge Kutta Method

Consider the differential equation $\frac{dy}{dx} = f(x,y)$ with $y_0 = y(x_0)$ and $x_1 = x_0 + h,\ x_2 = x_1 + h$.

Runge kutta formula of second order:

$y_1 = y_0 + k,\ where\ k = \frac{1}{2}(k_1 + k_2)$

$k_1 = hf(x_0, y_0)$

$k_2 = hf(x_0 + h, y_0 + k_1)$

Runge kutta formula of fourth order:

$$y_1 = y_0 + k, \qquad where\ k = \frac{1}{6}(k_1 + 2k_2 2k_3 + k_4)$$

$k_1 = hf(x_0, y_0)$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

Examples: Runge Kutta Method

1. Using Runge-Kutta method of second order find the value of y at $x = 0.1$ and $x = 0.2$. given that $\frac{dy}{dx} = x + y$ with $y(0) = 1$, $h = 0.1$. [Ans=1.11].
2. Using Runge-Kutta method of second order find the value of y at $x = 0.1$ given $y = 1.2$ when $x = 0$ and $\frac{dy}{dx} = 3x + y^2$, $h = 0.1$.
3. By using Runge-Kutta method of fourth order find the value of y at $x = 0.1$. given that $\frac{dy}{dx} = x + y^2$ with $y(0) = 1$ and $h = 0.1$.
4. By using Runge-Kutta method of fourth order find the value of y at $x = 1.2$. given that $\frac{dy}{dx} = xy$ with $y(1) = 2$ and $h = 0.2$.
5. By using Runge-Kutta method of fourth order find the value of y at $x = 0.2$. given that $\frac{dy}{dx} = \sqrt{x + y}$ with $y(0) = 1$ and $h = 0.2$. [Ans=1.2167].
6. By using Runge-Kutta method of fourth order find the value of y at $x = 0.2$. given that $\frac{dy}{dx} = \frac{1}{x+y}$, with $x_0 = 0, y_0 = 1$ and $h = 0.2$. [Ans=1.1697].
7. By using Runge-Kutta method of fourth order find the value of y at $x = 1.2$. given that $\frac{dy}{dx} = x^2 + y^2$ with $y(1) = 1.5$ and $h = 0.1$. [Ans=1.8955].

Examples

1 Using modified Euler's method solve equation $\frac{dy}{dx} = x - y^2$ with $y(0) = 1$ to calculate y at $x = 0.2$ take $h = 0.2$.

Solution: Let $f(x, y) = x - y^2$ and $x_0 = 0,\ y_0 = 1$

By Eulers method $y_1 = y_0 + hf(x_0, y_0) = 1 + 0.2(0 - 1^2) = 0.8$

By modified Eulers method $y_1^{(1)} = y_0 + hf\left(x_0, y_1^{(0)}\right) = 1 + 0.2(0 - (0.8)^2) = 0.872$

$y_1^{(2)} = y_0 + hf\left(x_0, y_1^{(1)}\right) = 1 + 0.2(0 - (0.872)^2) = 0.848$

$y_1^{(3)} = y_0 + hf\left(x_0, y_1^{(2)}\right) = 1 + 0.2(0 - (0.848)^2) = 0.856$

$y_1^{(4)} = y_0 + hf\left(x_0, y_1^{(3)}\right) = 1 + 0.2(0 - (0.856)^2) = 0.853$

2 Using modified Euler's method, find the value of y at $x = 1.2$ correct up to three decimal places given that, $\frac{dy}{dx} = log(x + y)$ with $y(1) = 2, h = 0.2$.

Solution: Let $\frac{dy}{dx} = log(x + y)$ and $x_0 = 1,\ y_0 = 2$

By Eulers method $y_1 = y_0 + hf(x_0, y_0) = 2 + 0.2\ log(1 + 2) = 2.0954242$

By modified Eulers method $y_1^{(1)} = y_0 + h\ f\left(x_0, y_1^{(0)}\right) = 2 + 0.2\ log(1 + 2.09542) = 2.09814$

$y_1^{(2)} = y_0 + h\ f\left(x_0, y_1^{(1)}\right) = 2 + 0.2\ log(1 + 2.09814) = 2.09822$

$y_1^{(3)} = y_0 + h\ f\left(x_0, y_1^{(2)}\right) = 2 + 0.2\ log(1 + 2.09822) = 2.098222$

$y_1^{(4)} = y_0 + h\ f\left(x_0, y_1^{(3)}\right) = 2 + 0.2\ log(1 + 2.098222) = 2.098222$

3 Using modified Euler's method, find the value of y at $x = 0.1$, given $\frac{dy}{dx} = x^2 + y$ with $y(0) = 1$,
$h = 0.1$.

Solution: Let $\frac{dy}{dx} = x^2 + y$ and $x_0 = 0,\ y_0 = 1$

By Eulers method $y_1 = y_0 + hf(x_0, y_0) = 1 + 0.1(0 + 1) = 1.1$

By modified Eulers method $y_1^{(1)} = y_0 + h\ f\left(x_0, y_1^{(0)}\right) = 1 + 0.1(0 + 1.1) = 1.11$

$y_1^{(2)} = y_0 + h\ f\left(x_0, y_1^{(1)}\right) = 1 + 0.1(0 + 1.11) = 1.111$

$y_1^{(3)} = y_0 + h\ f\left(x_0, y_1^{(2)}\right) = 1 + 0.1(0 + 1.111) = 1.1111$

$y_1^{(4)} = y_0 + h\ f\left(x_0, y_1^{(3)}\right) = 1 + 0.1(0 + 1.1111) = 1.11111$

4 Apply fourth order Runge Kutta method to find value of y at $x = 0.2$ for the differential equation $\frac{dy}{dx} = \sqrt{x + y}$ subject to the conditions $y = 1$ for $x = 0$ and $h = 0.2$.

Solution: Given $\frac{dy}{dx} = \sqrt{x + y},\ x_0 = 0,\ y_0 = 1,\ h = 0.2\ and\ x_1 = x_0 + h = 0.2$

By Runge Kutta fourth order method

$k_1 = h\ f(x_0, y_0) = (0.2)1 = 0.2$

$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.2\left[\sqrt{0.1 + 1.1}\right] = 0.219089$

$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.2\left[\sqrt{0.1 + 1.109544}\right] = 0.21995854$

$k_4 = hf(x_0 + h, y_0 + k_3) = 0.2\left[\sqrt{0.2 + 1.21995854}\right] = 0.2383240$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.2194038$

$y_1 = y_0 + k = 1.2194038.$

5 Apply fourth order Runge Kutta method to find value of y at $x = 0.1$ for the differential equation $\frac{dy}{dx} = \frac{1}{x+y}$ subject to the conditions, $y = 1$ for $x = 0$ and $h = 0.1$.

Solution: Given $\frac{dy}{dx} = \frac{1}{x+y}$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$ *and* $x_1 = x_0 + h = 0.1$

By Runge Kutta fourth order method

$k_1 = h\,f(x_0, y_0) = (0.1)1 = 0.1$

$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1\left[\frac{1}{0.05 + 1.05}\right] = 0.090909$

$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1\left[\frac{1}{0.05 + 1.0454545}\right] = 0.09128$

$k_4 = hf(x_0 + h, y_0 + k_3) = 0.1\left[\frac{1}{0.1 + 1.09128}\right] = 0.839433$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.217302$

$y_1 = y_0 + k = 1 + 0.217302 = 1.217302.$

6 Apply fourth order Runge Kutta method to find value of y at $x = 0.1$ for the differential equation $\frac{dy}{dx} = x^2 + y^2$ subject to the conditions $y = 1$ for $x = 0$ and $h = 0.1$.

Solution: Given $\frac{dy}{dx} = x^2 + y^2$ $x_0 = 0$, $y_0 = 1$, $h = 0.1$ *and* $x_1 = x_0 + h = 0.1$

By Runge Kutta fourth order method

$k_1 = h\,f(x_0, y_0) = (0.1)1 = 0.1$

$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1[0.05^2 + 1.05^2] = 0.1105$

$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1[0.05^2 + 1.05525^2] = 0.11160$

$k_4 = hf(x_0 + h, y_0 + k_3) = 0.1[0.1^2 + 1.055822^2] = 0.1124$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.109433$

$y_1 = y_0 + k = 1.109433.$